

Requirements Table

Lab 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #44 – Database Query Performance Profiler

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall parse Postgres and MySQL EXPLAIN plans, detect full table scans on tables with more than 10,000 rows, and suggest indexes to fix them.	High	Given a query with a seq scan, the tool flags it and gives an index recommendation within 3 seconds. Fails if a bad query is marked as fine.	Fixing seq scans is the whole point of this tool.	Given in problem statement.
FR-002	Functional	The system shall accept slow query log files or live-streamed logs from a database agent, and store all entries in a format that can be searched.	High	Upload 10,000 entries, all must appear in the system within 2 minutes. Each record needs timestamp, query text and duration. Fail if anything is missing.	Every other feature needs this data to exist.	Should work for both file upload and live agent.
FR-003	Functional	The system shall go through ingested queries and find which columns are missing indexes based on their usage in WHERE, JOIN and ORDER BY clauses.	High	On a test set with known answers, should catch at least 90% of missing indexes. Fails if it catches less than 70% or has too many false positives (>20%).	DBAs do this manually and it takes hours. This automates it.	Thresholds based on what's reasonable for the dataset size.
FR-004	Functional	The system shall email a weekly report every Monday with the 10 slowest queries, index recommendations and a comparison against last week.	Medium	Report arrives by 8 AM Monday with all three sections present. Fails if it's more than 1 hour late or a section is blank.	Easier to track improvements over time if you get a regular summary.	Schedule should be configurable.
FR-005	Functional	The system shall show a live dashboard of slow queries happening right now, with their plans and suggested fixes. Filterable by database, time and severity.	Medium	New slow queries show up within 5 seconds. All filter combos give correct results. Fails if delay is over 30 seconds or filters break.	Useful when issues pop up during peak hours and you can't wait for Monday.	Should work on a regular laptop.
NFR-001	Non-Functional	The system shall handle up to 5,000 slow query log entries per minute without dropping records or slowing down the database it's monitoring.	High	10-minute load test at 5,000 rec/min: zero records lost, host DB CPU stays under 5% extra load. Fails if any record is dropped.	The tool can't be the thing that slows the DB down.	Given in problem statement.
NFR-002	Non-Functional	The system shall use role-based access control so only DBAs and Backend Leads can see query logs and recommendations.	High	Any user without the right role gets a 403 on protected pages. All access is logged with who and when. Fails if unauthorised user gets data.	Slow query logs can expose internal app logic and sometimes credentials.	Should support SSO.